SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	Manoj M	**Reg. No.:**	192511060
Section / Batch:	CSE	**Date:**	29.07.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

Q51. A smart grid system collects power consumption readings that arrive in random order and must be sorted periodically for analysis.
a) Analyze how Selection Sort and Bubble Sort perform on this dataset.
b) Compare their efficiency in terms of comparisons and swaps.
c) Recommend the most suitable algorithm with justification.

Code of Conduct

I _Manoj(192511060)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

RUBRICS FOR CALCULATION

Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario

Application of Concepts	30%	Applies appropriate concepts effectively to solve complex realworld problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply concepts but appropriate concepts
Analytical & DecisionMaking Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Faculty In-charge	Course Coordinator	Head of Department
Name:Name:Name:		
Signature: ________________	________________Signature:	________________Signature:
Date: ____________________	Date: ____________________	Date: ____________________

W

a) Analyze how Selection Sort and Bubble Sort perform on this dataset.

5 1) smart grid system - selection sort vs Bubble sort

1. Analytic problem solving

★ A small grid system receives power consumption reading in random order. Before Analysis, the Reading must be sorted in ascending order.

Selection sort

- Finds the Minimum element in the unsorted part of an array.
- Places it in its correct after each pass.
- Number of comparision is $n(n-1)/2$
- Performs only one swap per pass, making it suitable when swapping is expensive.

Bubble sort

- compares adjacent elements and swaps them if they are in the wrong order.
- Largest element it into correct position after each pass.
- Performs many swaps when the data is randomly arranged.

Recommendations

* It performs significantly fewer swaps
* It gives predictable performance
* Bubble sort performs excessive swaps on random dataset, increasing execution time.

2.Scenario-Based – Execution

Algorithm Used: Selection Sort

Algorithm

1. Read all power consumption readings.
2. Start from the first element.
3. Find the smallest value in the remaining unsorted array.
4. Swap it with the current position.
5. Repeat until all readings are sorted.
6. Display the sorted readings.

Example

Input

Pass 1

Pass 2

Pass 3

Pass 4

Output

3.Debugging & Optimisation Task – Execution

Buggy Code

Issues Identified

1. Wrong comparison operator (>).
2. The code selects the largest element instead of the smallest.
3. Output becomes sorted in descending order instead of ascending order.

Corrected Code

Optimisation

- Swap only when the minimum element changes (min != i).
- Reduces unnecessary swaps.
- Maintains **$O(n^2)$** time complexity while improving practical performance.

Expected Output

INPUT /OUTPUT:

Input

ARR[]={14,3,2,5,4,5,6,8,9,97,90,454,5455,21212,2,424,4,43,4343}

Output:

SELECTION SORT:

ARR[]={2,3,4,5,6,6,8,9,14,90,97,454,5455,424,43,4343}